

Missouri University of Science and Technology

Intro to Data Science
CS 3402 (Spring 2026)

Lecture 5 Descriptive Statistics I

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Intro to Descriptive Statistics

Descriptive statistics helps to simplify large amounts of data in a sensible way.

Present quantitative descriptions of data. It is simply a way to describe the data.

- a *population* is a collection of objects, items ("units") about which information is sought;

- a *sample* is a part of the population that is observed.

Population and Sample.

A city has **12,000 residential homes**.
A real-estate analyst wants to estimate the **average price per square foot** of homes in the city.

Instead of collecting data from all homes, the analyst randomly selects **300 homes** and records their price per square foot.

Questions:
1. What is the **population** in this study?
2. What is the **sample** in this study?
3. How many **units** are in the population?
4. How many **units** are in the sample?

Descriptive statistics

Describe the basic features of the samples in a study.

Provide summaries about the samples as an approximation of the population.

Simple graphics (Visualization).

Descriptive statistics - Steps

- *Data Preparation*: Given a specific project, we need to prepare the data for generating statistically valid description.

- *Descriptive statistics*: Different statistics to describe and summarize the data concisely and evaluate different ways to visualize them.

Data Preparation – Part 1

Collect and prepare in a format appropriate for analysis of the samples.

How to collect/prepare data?

- *Obtaining the data*: data can be read from files/ obtained by scraping the web
- *Parsing the data*: parsing procedure depends on what format the data are in: plain text, fixed columns, CSV, XML, HTML, etc.

Data Preparation – Part 2

Collect and prepare in a format appropriate for analysis of the samples.

How to collect/prepare data?

- *Cleaning the data*: Survey responses and other data files are almost always incomplete. How to deal with incomplete data?

- *Building data structures*: store them in a data structure that lends itself to the analysis we are interested in. What data structure?

Data Preparation – The Adult Dataset

"Adult" Dataset from UCI's Machine Learning Repository

Observations concerning different financial parameters related to the US population: age, sex, marital (marital status of the individual), country, income (Boolean variable: whether the person makes more than $50,000 per annum), education (the highest level of education achieved by the individual), occupation, capital gain, etc

Data Preparation – The Adult Dataset

"Adult" Dataset from UCI's Machine Learning Repository

URL: https://archive.ics.uci.edu/dataset/2/adult

Please go to the website to see details.

we can explore the data by asking questions like: "*Are men more likely to become high-income professionals than women, i.e., to receive an income of over $50,000 per annum?*"

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1) Understand the Dataset

sex — Male or Female

income — Two categories: >50K (high income) or <=50K (not high income)

Data Preparation – The Adult Dataset

URL: https://archive.ics.uci.edu/dataset/2/adult

we can explore the data by asking questions like: "*Are men more likely to become high-income professionals than women, i.e., to receive an income of over $50,000 per annum?*"

2) Formulate the Hypothesis

Null Hypothesis — There is no difference in the probability of earning >50K between men and women.

Alternative Hypothesis — Men are more likely than women to earn >50K.

Data Preparation – The Adult Dataset

URL: https://archive.ics.uci.edu/dataset/2/adult

we can explore the data by asking questions like: *"Are men more likely to become high-income professionals than women, i.e., to receive an income of over $50,000 per annum?"*

3) Compute the Observed Frequencies

- Count how many men and women earn >50K

```
male_high = df[(df['sex']=='Male') & (df['income']=='>50K')].shape[0]
female_high = df[(df['sex']=='Female') & (df['income']=='>50K')].shape[0]
male_total = df[df['sex']=='Male'].shape[0]
female_total = df[df['sex']=='Female'].shape[0]
```

- Calculate proportions.

```
p_male = male_high / male_total
p_female = female_high / female_total
```

Data Preparation – The Adult Dataset

we can explore the data by asking questions like: *"Are men more likely to become high-income professionals than women, i.e., to receive an income of over $50,000 per annum?"*

4) Perform a Statistical Test

```
from statsmodels.stats.proportion import
proportions_ztest

count = np.array([b, d])      # high income counts
nobs = np.array([m, f])       # total counts
zstat, pval = proportions_ztest(count, nobs)
```

Sex	<=50K	>50K	Total
Male	a	b	m
Female	c	d	f
Total			n

Data Preparation – The Adult Dataset

5) Interpretation Example

Suppose after counting:

5,000 of 20,000 men earn >50K → 25%

1,500 of 15,000 women earn >50K → 10%

Men's proportion is higher than women's.

A statistical test (e.g., z-test) might give a very small p-value → significant difference.

Data Preparation – The Adult Dataset

6) Extensions

Suppose after counting:

5,000 of 20,000 men earn >50K → 25%

1,500 of 15,000 women earn >50K → 10%

Men are more likely than women to earn >$50,000 per year.

This analysis **does not prove causation** — many factors (education, hours worked, occupation) affect income.

income = coefficient1 * sex + coefficient2 * age + coefficient3 * education + coefficient4 * hours_per_week + ...

Exploratory Data Analysis

- Mean/Average

$$\mu = \frac{1}{n} \sum_{i=1}^{n} x_i .$$

where n is the number of samples, xi is the values.

Exploratory Data Analysis

- Sample Variance

$$\sigma^2 = \frac{1}{n}\sum_i (x_i - \mu)^2.$$

where n is the number of samples, xi is the values, and u is the mean.

Standard deviation (STD)?

Exploratory Data Analysis

- Sample Median

Median is an order statistic giving the middle value of a sample.

All the values are ordered by their magnitude and the median is defined as the value that is **in the middle of the ordered list**.

[34, 12, 1, 10000000, 46, 98, 3, 56, 87] [34, 12, 1, 10000000, 46, 98, 3, 56]

An item of data that is significantly different from the rest of the data is called an outlier.

Median is much more robust in the face of outliers.

Exploratory Data Analysis

- Quantiles and Percentiles

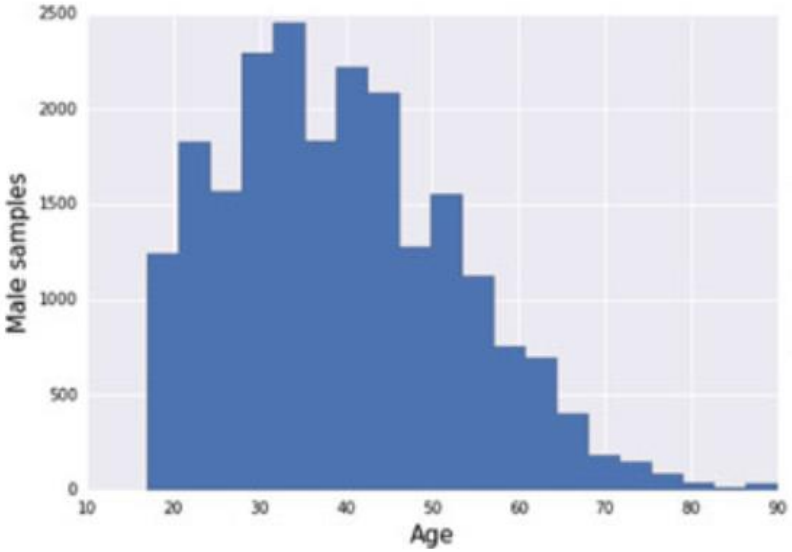
Measures how sample data are distributed in general.

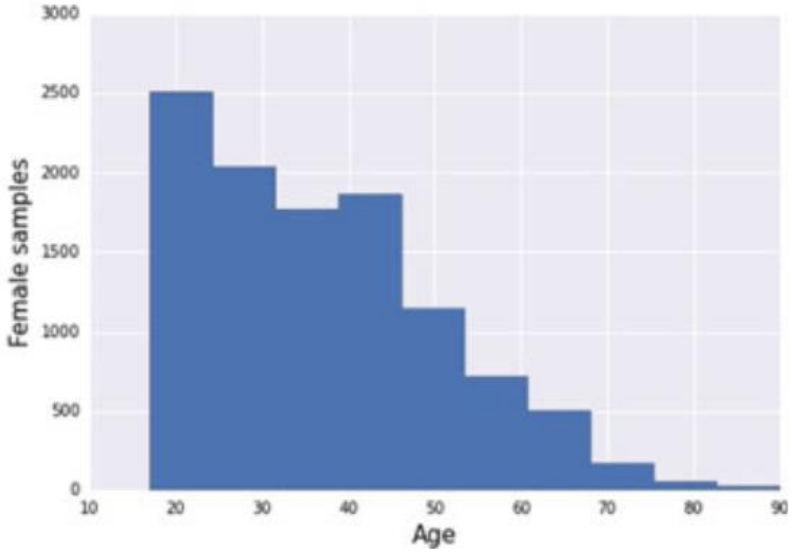
order the samples {xi}, then find the x_p so that it divides the data into two parts:
- a fraction p of the data values is less than or equal to x_p
- the remaining fraction .(1 − p) is greater than x_p.

x_p is the pth quantile, or the 100 × pth percentile

Data Distribution

histogram, a graph that shows the frequency of each value.





Data Distribution

The Cumulative Distribution Function (CDF): the probability that a real-valued random variable X with a given probability distribution will be found to have a value less than or equal to x.

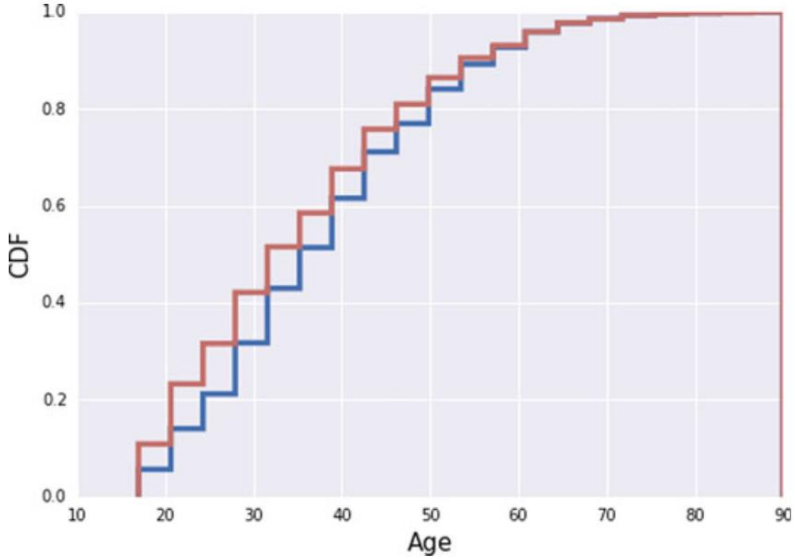


Fig. 3.3 The CDF of the age of working male (in blue) and female (in red) samples

Outlier Treatment

Outliers are data samples with a value that is far from the central tendency.

Different rules can be defined to **detect outliers**,

- Computing samples that are far from the median.
- Computing samples whose values exceed the mean by 2 or 3 standard deviations.

Option 1: Keep them
Use when:
Values are valid
Outliers represent real variability

Option 2: Remove them
Appropriate when:
You can **justify** removal (measurement or data-entry error)

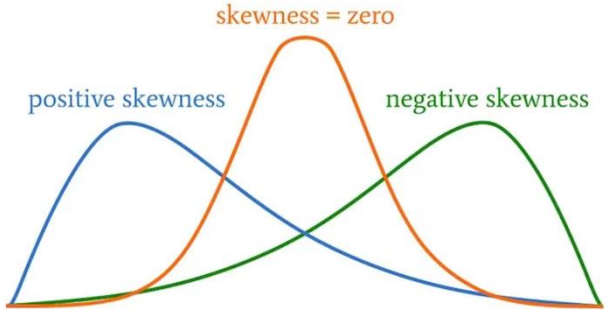
Measuring Asymmetry: Skewness

Skewness is a statistic that measures the asymmetry of the set of .n data samples

$$g_1 = \frac{1}{n} \frac{\sum_i (x_i - \mu^3)}{\sigma^3},$$

where μ is the mean, σ is the standard deviation, and n is the number of data points.

Negative deviation indicates that the distribution "skews left" (it extends further to the left than to the right).



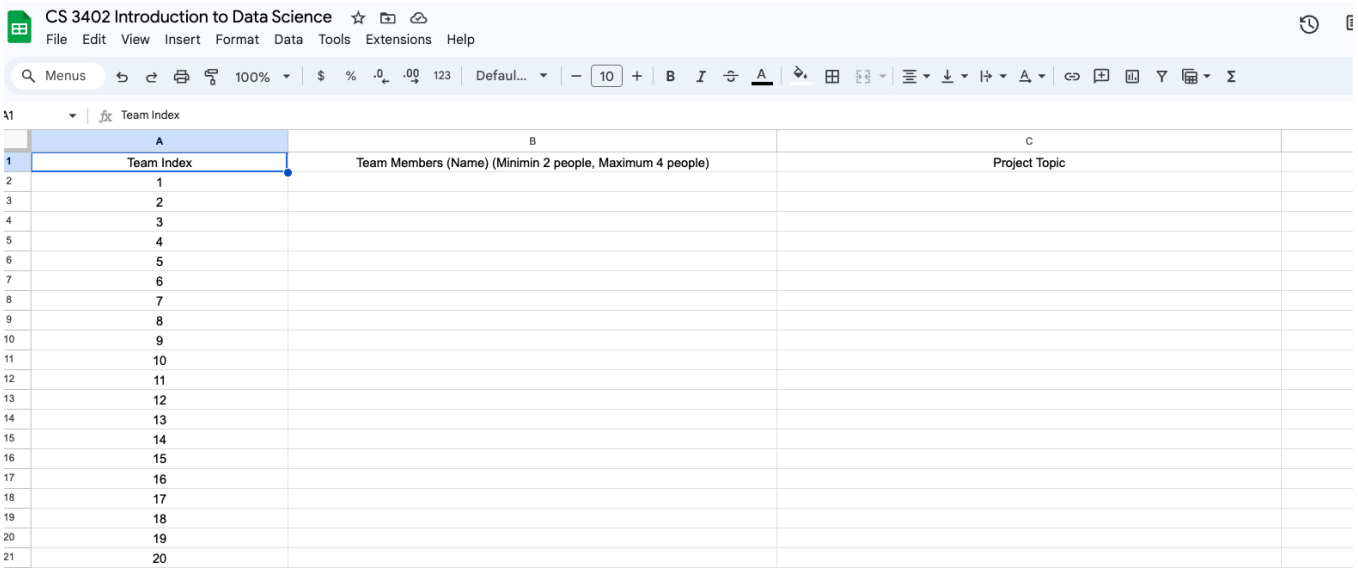
Notes

Group Project Registration:

https://docs.google.com/spreadsheets/d/12lqUdHOODhmKNLq16DMRyuNSlyHW77cPyDtfucWWbWU/edit?usp=sharing

1. You need to use your MST email to access and edit the sheet.

2. Leave project topic blank (before intro to project topics).



	A	B	C
1	Team Index	Team Members (Name) (Minimin 2 people, Maximum 4 people)	Project Topic
2	1		
3	2		
4	3		
5	4		
6	5		
7	6		
8	7		
9	8		
10	9		
11	10		
12	11		
13	12		
14	13		
15	14		
16	15		
17	16		
18	17		
19	18		
20	19		
21	20		

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Feel free to find teammates and build your teams.

Otherwise, I will randomly assign teams of 2–4 people.

DDL: 02/15/2026

Homework 1 – Search/Collect/Process Data

Problem Definition: Comparing the median listing price per square foot in Rolla, MO, using **official data** versus **scraped data**.

Background:

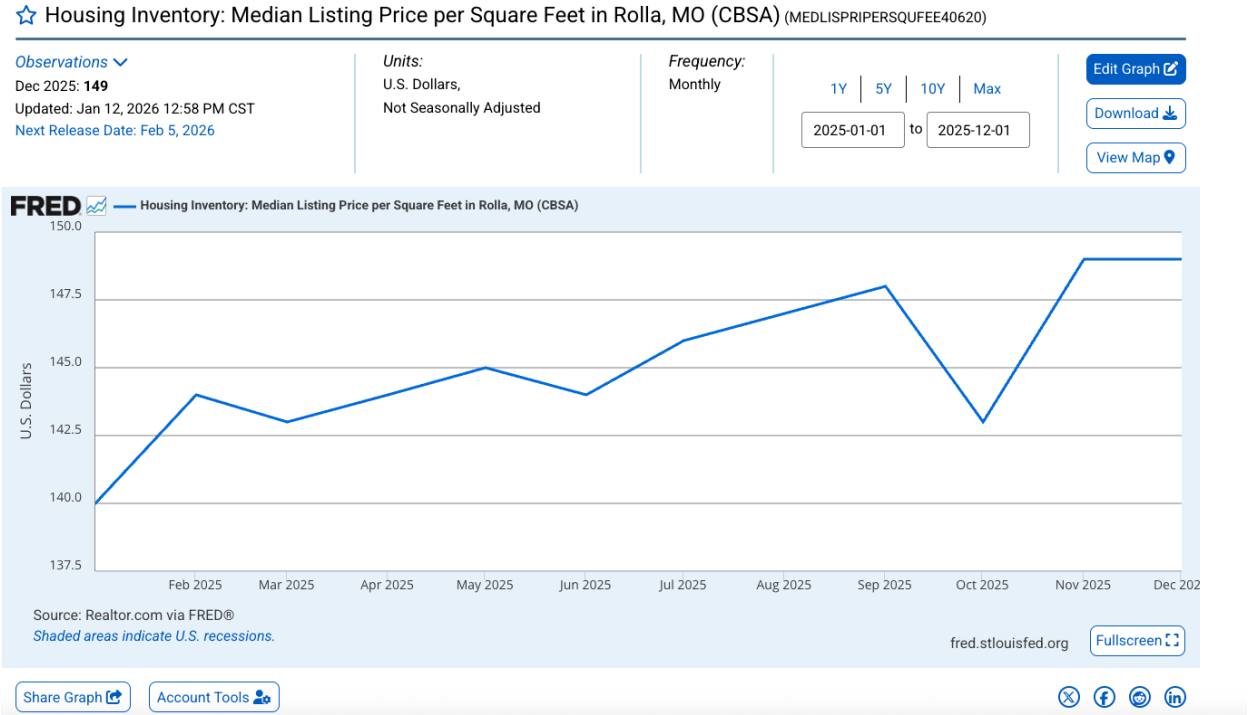
1. The Federal Reserve Economic Data (FRED) provides an official estimate of the **median listing price per square foot** for Rolla, MO (MSA 40620):

   FRED series: *Median Listing Price per Square Foot*
   https://fred.stlouisfed.org/series/MEDLISPRIPERSQUFEE40620

2. Real-estate websites (e.g., Zillow, Redfin, Realtor.com) also report listing prices, but their estimates may differ due to data coverage, methodology, and delayed data.

Homework 1 – Search/Collect/Process Data

For official data: Get the latest **median** listing price per square feet in Rolla

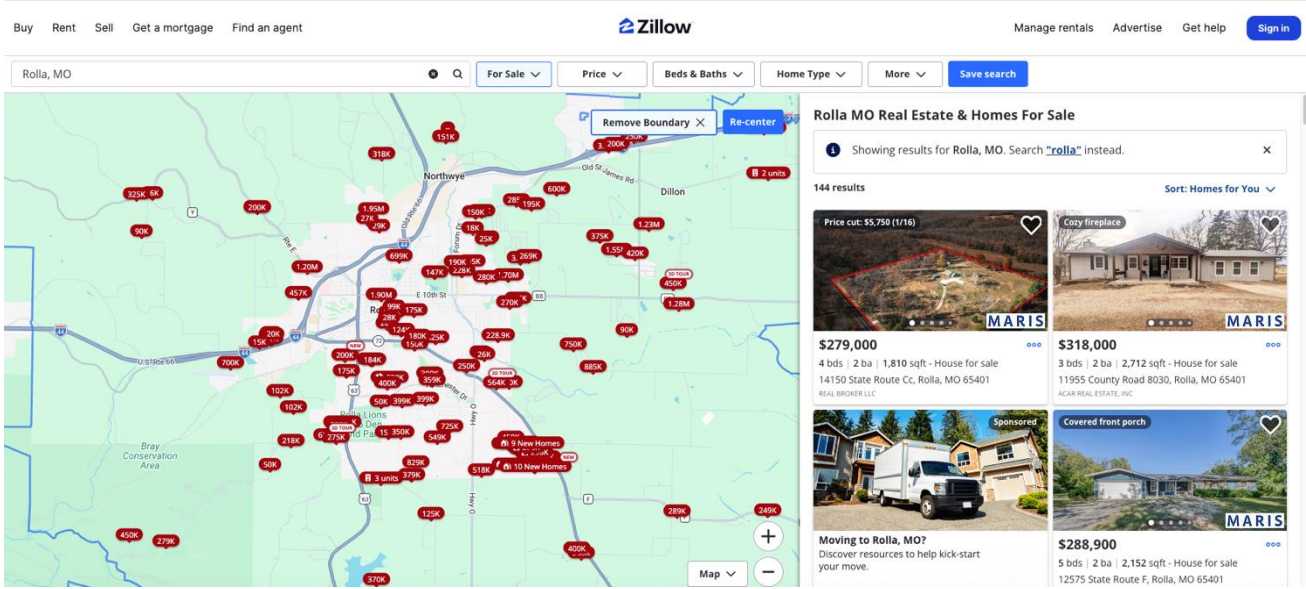


Source: https://fred.stlouisfed.org/series/MEDLISPRIPERSQUFEE40620

Homework 1 – Search/Collect/Process Data

For Real-estate websites:

1. Select **one real-estate website** (e.g., Zillow, Redfin, Realtor.com).
2. Collect data for **active house listings in Rolla, MO**, including: Listing price and Square footage
3. Compute the **price per square foot** for each listing: Price per sq ft = Listing price/Square feet
4. Calculate the **median price per square foot** from your collected data.



Homework 1 – Search/Collect/Process Data

For Real-estate websites:



Example data samples (You need to search and collect data by yourself!)

Sample ID	House Price	Square Feet
1	89, 900	600
2	300, 300	1,576
3	379, 000	3,882
4	1,700,000	11,396

Data sources: realtor.com

1. Use Python to get house price per square feet
2. Write python code and use matplotlib to visualize the price and square feet in a plot
3. Get **median price** per square feet in python

Homework 1 – Search/Collect/Process Data

Problem Definition: Comparing Median Listing Price per Square Feet (Rolla, MO). (official-data vs. scraped-data)

Last step:

Diff(median price per square feet from FRED, median price per square feet from real-estate websites,)

What metrics can be used here?

Mean square error, Absolute Difference, Relative Difference, Ratio, etc....

Submission requirement: Upload your python code file with detailed comments.

AI Tools can be used, but their use **must be clearly disclosed** in your submission (e.g., which tools were used and for what purpose).

Will create an assignment on Canvas. DDL: 02/22/2026

Notes

We need volunteers to present Python/Timm/PyTorch/IDEs. (send me you name and MST ID)

Will upload the project/homework to Canvas.

TA: Jiaqian Zhu (jzbmn@mst.edu) Office hours: 1:00 – 2:00 pm on Mondays/Wednesdays

PyTorch & IDE: Kaiser, Luke; Chenevert, William (Jan. 29th)

Python & Timm & Huggingface: Lewis, James (Jan. 27th)